

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – Transformer Mechanics Worksheet

|                  |                                                                                                                                                                       |               |                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------|
| Course Code:     | CSA6502                                                                                                                                                               | Course Title: | Generative AI and Large Scale Model                       |
| CO Assessed:     | CO1-Imitation (BL3)                                                                                                                                                   | Assessment:   | Assessment Tool 3– <b>Transformer Mechanics Worksheet</b> |
| Weightage:       | 10% of CIA                                                                                                                                                            | Total Marks:  | 20                                                        |
| CO1 Statement:   | Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. |               |                                                           |
| Student Name:    | CHAVALI CHANDU                                                                                                                                                        | Reg. No.:     | 192472332                                                 |
| Section / Batch: | SLOT A                                                                                                                                                                | Date:         |                                                           |

### Code of Conduct

I 192472332/CHAVALI CHANDU(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

### Instructions

- This assessment contains 4 set of questions.
- Each student assigned with one set of questions .
- Each question carries 5 marks. Total: 20 Marks.
- Ensure clarity, structure, and logical flow in diagrams.
- Duration: 20 Minutes.

### Annexure A Transformer Mechanics Worksheet

## Transformer Mechanics Worksheet — SET A

**Question 1:** Tokenization (5 marks). The word "UNHAPPINESS" is uncommon enough that a sub-word tokenizer would likely split it into smaller pieces rather than keep it whole.

(a) Split "UNHAPPINESS" into 3 reasonable sub-word pieces (there's no single "correct" answer — justify your split).

**(a) Answer**

**Split "UNHAPPINESS" into 3 reasonable sub-word pieces**

A reasonable split is:

**UN + HAPPY + NESS**

**Justification:**

- **UN** → Prefix meaning *not* or *opposite of*.
- **HAPPY** → Root word representing the main meaning.
- **NESS** → Suffix that changes the adjective *happy* into the noun *happiness*.

This split is meaningful because each sub-word is a common prefix, root word, or suffix that sub-word tokenizers (such as WordPiece or BPE) are likely to recognize and reuse across many different words.

**Answer:**

**UNHAPPINESS → UN + HAPPY + NESS**

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(b) In 1-2 sentences, explain WHY sub-word tokenization (rather than whole-word tokenization) helps a model handle a word like this one, even if the model has never seen the whole word before in training.

**(b) Answer**

Sub-word tokenization divides a word into smaller meaningful units, such as prefixes, root words, and suffixes, instead of treating the entire word as a single token. For example, "UNHAPPINESS" can be split into "UN" + "HAPPY" + "NESS", where each sub-word has its own meaning. Even if the model has never encountered the complete word "UNHAPPINESS" during training, it has likely learned these individual sub-word units from other words. By combining the meanings of these known pieces, the model can infer that "UNHAPPINESS" refers to the state of not being happy. This approach reduces the **Out-of-Vocabulary (OOV)** problem, decreases the vocabulary size, improves the model's ability to process rare or newly formed words, and enhances overall language understanding and prediction accuracy.

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**Question 2:** Embeddings (5 marks). Consider the word pairs "cat" and "kitten" and "cat" and "car".

(a) Which pair would you expect to have HIGHER cosine similarity in embedding space,

and why?

**Answer:**

The pair **"cat"** and **"kitten"** would have a **higher cosine similarity** than **"cat"** and **"car"** because **cat** and **kitten** are semantically related. A **kitten** is a young **cat**, so both words frequently appear in similar contexts, such as discussions about pets, animals, and veterinary care. As a result, their word embeddings are located close to each other in the embedding space, resulting in a **high cosine similarity**.

In contrast, **cat** and **car** are unrelated words. A **cat** refers to an animal, whereas a **car** is a vehicle. Since they occur in very different contexts and have different meanings, their embeddings are farther apart in the vector space, resulting in a **lower cosine similarity**.

**Therefore, the pair with the higher cosine similarity is:**

**Cat ↔ Kitten**

because they share similar meanings and contexts, while **Cat ↔ Car** represents unrelated concepts and has a much lower similarity.

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(b) In your own words, what does an embedding actually represent, and how is it different from a one-hot vector?

**Answer:**

An **embedding** is a dense numerical vector that represents the meaning and context of a word in a continuous vector space. Words with similar meanings or that appear in similar contexts, such as **"cat"** and **"kitten"**, have embedding vectors that are close to each other. These vectors are learned automatically during model training and capture semantic and syntactic relationships between words.

In contrast, a **one-hot vector** is a sparse representation in which only one element is **1** and all the remaining elements are **0**. Each word has a unique position in the vector, but one-hot vectors do not contain any information about the meaning or relationship between words. For example, the one-hot vectors for **"cat"** and **"kitten"** are just as different as those for **"cat"** and **"car"**, even though **"cat"** and **"kitten"** are semantically related.

Therefore, **embeddings are more effective than one-hot vectors** because they capture the semantic meaning, contextual similarity, and relationships between words while also reducing dimensionality and improving the performance of Natural Language Processing (NLP) models.

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**Question 3:** Self-Attention Worked Example (5 marks). Read the sentence: "The trophy didn't fit in the suitcase because it was too big."

(a) The word "it" should attend most strongly to which other word in the sentence? Circle one: "trophy" / "suitcase"

**Answer:**

✔ Circle: trophy

**Explanation:**

In this sentence, the pronoun "it" refers to the "trophy" because the phrase "was too big" explains why the trophy could not fit inside the suitcase. A trophy that is too big would prevent it from fitting into the suitcase. Therefore, during self-attention, the word "it" attends most strongly to "trophy", allowing the model to correctly understand the meaning of the sentence.

**Final Answer:** ○ Trophy

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(b) Explain in 2-3 sentences WHY self-attention (not a simple word-order rule) is needed to correctly resolve what "it" refers to here.

**Answer:**

Self-attention helps the model determine that "it" refers to the "trophy" by considering the meaning and relationship between all the words in the sentence, rather than relying only on their positions. A simple word-order rule might incorrectly assume that "it" refers to the nearest noun ("suitcase"), but self-attention understands that the trophy is the object that is too big to fit inside the suitcase. This enables the model to correctly capture the context and resolve the reference accurately.

**Question 4:** Query, Key, Value (5 marks). In your own words, briefly explain what the Query, Key, and Value vectors each represent in self-attention, and how they combine to produce an attention score.

**Answer:**

In the **self-attention mechanism**, every word in a sentence is converted into three vectors: **Query (Q)**, **Key (K)**, and **Value (V)**. These vectors help the model determine which words are important and how much attention each word should receive.

- **Query (Q):** Represents what the current word is looking for or the information it needs from other words in the sentence.
- **Key (K):** Represents the characteristics or features of each word that can be matched with a query.
- **Value (V):** Contains the actual information or representation of each word that will be passed to the output if that word receives attention.

The attention score is calculated by comparing the **Query** vector of one word with the **Key** vectors of all other words using a similarity measure, such as the dot product. A higher similarity indicates that the words are more closely related, resulting in a higher attention score. These scores are then normalized using the **Softmax** function to produce attention weights. Finally, the weighted sum of the **Value** vectors is computed to generate the output representation, allowing the model to focus on the most relevant words while understanding the context of the sentence.

**In summary:**

- **Query (Q):** What the word is searching for.
  - **Key (K):** What each word offers for matching.
  - **Value (V):** The actual information passed to the output.
  - **Attention Score:** Obtained by comparing Query and Key vectors; the resulting weights are applied to the Value vectors to produce the final contextual representa
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## Rubrics for Calculation

| Criteria                             | Wt. | Level 4 (Exemplary)                                                                                  | Level 3 (Proficient)                              | Level 2 (Developing)                                             | Level 1 (Beginning)                             |
|--------------------------------------|-----|------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------|
| <b>Technical / Concept</b>           | 30% | Accurately explains tokenization, embeddings, and self-attention mechanics using correct terminology | Explains most mechanics correctly with minor gaps | Partial or superficial understanding of the mechanics            | Major misconceptions about core concepts        |
| <b>Application</b>                   | 25% | Correctly traces a worked example (e.g. computing attention relevance) step-by-step                  | Traces most steps correctly with minor errors     | Significant errors when applying mechanics to the worked example | Cannot trace or apply mechanics at all          |
| <b>Quality</b>                       | 20% | Clear, logical, mathematically sound reasoning throughout                                            | Mostly sound reasoning with minor errors          | Reasoning has notable gaps or inconsistencies                    | Reasoning is disorganized and largely incorrect |
| <b>Presentation</b>                  | 15% | Neat, well-labeled diagrams/working, easy to follow                                                  | Reasonably clear presentation of working          | Presentation is unclear in places                                | Difficult to read presentation of working       |
| <b>Documentation / Communication</b> | 10% | Shows all work with clear justification for every step                                               | Shows most work with adequate justification       | Minimal work shown                                               | No work shown or answers are unjustified        |